How LLMs and Agents Change Research Pipelines and the Classroom

Milan Miric
University of Southern California,
Marshall School of Business

IE University

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Motivation

- **Textual Data in Research:** Text (10-Ks, hearings, transcripts) provides rich empirical signal for research questions that were previously difficult to measure.
- **The Data Bottleneck:** Existing methods (topic models, embeddings) require short documents and break down with long, noisy, boilerplate-heavy text.
- **Need for Control:** Unassisted, end-to-end LLM processing lacks auditability, transparency, and verifiable statistical guarantees.
- **Our Approach:** Build structured, modular AI pipelines. Use LLM to solve key bottlenecks (LLM text distillation) while using established econometric and empirical standards.

What LLMs Do

- **Two Core Strengths of LLMs:**
  - **Synthesizing text:** Converting long, unstructured text into concise, informative summaries.
  - **Generating code:** Writing structured instructions that can be systematically tested and audited.

- **Complementary Deployment:** Use LLMs at discrete bottlenecks where they excel, while retaining explicit control over data structures and validation.

- **Transparency & Replication:** Preserving intermediate chunks, prompts, and verification logs enables systematic replication. You can look and audit the code yourselves.

An Example: Privacy Regulation and Firm Innovation

- **Substantive Question:** How do government debates on consumer privacy regulation influence corporate strategy and innovation?
- **The Empirical Challenge:** Policy deliberations are diffuse across hundreds of hearing pages, while corporate responses are embedded in 10-K filings. Information is there, but Noise $>$ Signal.
- **Three-Stage Solution:**
  - **Step 1:** Summarize multi-hour Congressional hearings into dense policy summaries.
  - **Step 2:** Extract debated topics and directional member/witness stances.
  - **Step 3:** Connect hearing topics to S&P 500 10-K Item 1 business disclosures.

Part A

Engineering the AI-Enabled Research Pipeline

Step 1: Document Characteristics and Noise

- **Corpus Scale:** 2,352 Congressional hearings (1996–2025; House Science, Space, & Tech; Senate Commerce, Science, & Transportation).
- **Extreme Length:** Median transcript contains 36,505 words.
- **Running Example:** *Facebook, Social Media Privacy, and Data Abuse* (April 10, 2018): 243,434 words (385 PDF pages).
- **The Noise Problem:** Saturated with opening procedural preamble, committee pleasantries, and index dot leaders.

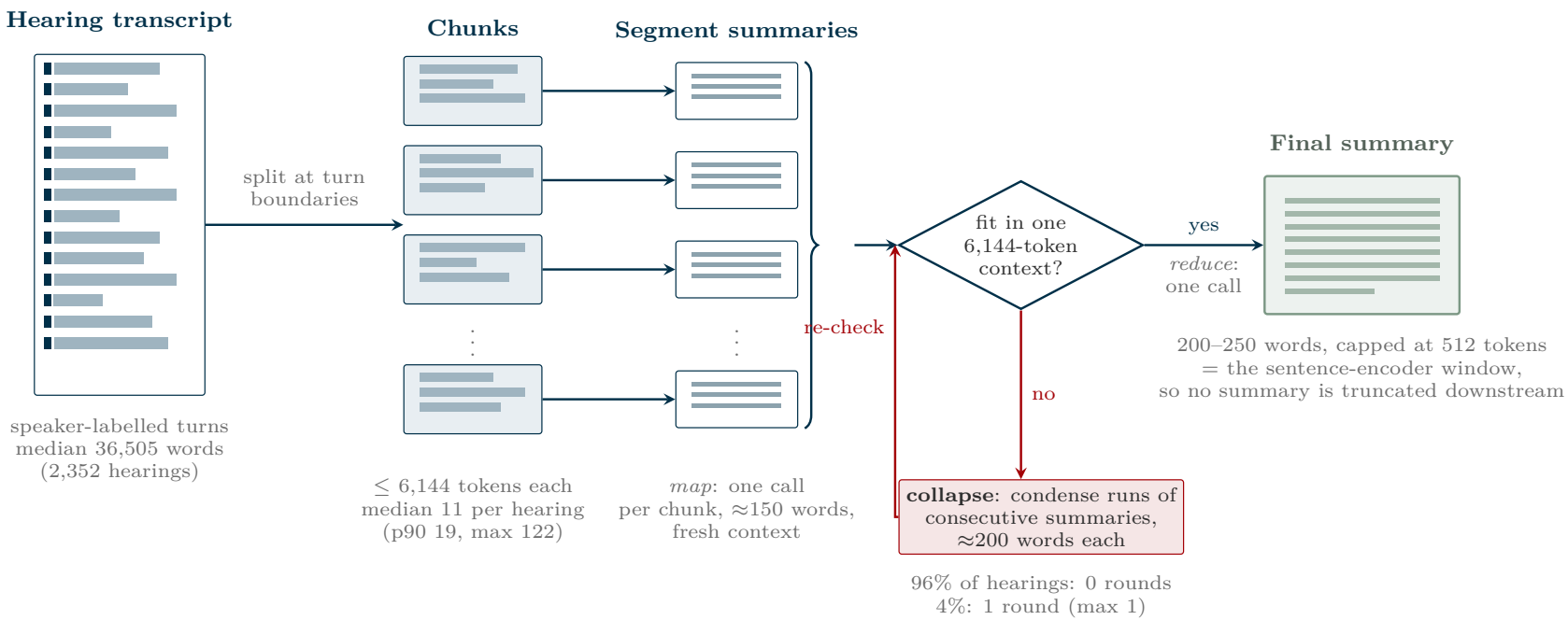
Raw Transcript Preamble Excerpt

Unidentified speaker: - FACEBOOK, SOCIAL MEDIA PRIVACY, AND THE USE AND ABUSE OF DATA [Senate Hearing 115-683]
[From the U.S. Government Publishing Office] — S. Hrg. 115-683
JOINT HEARING BEFORE THE COMMITTEE ON COMMERCE, SCIENCE, AND TRANSPORTATION
UNITED STATES SENATE AND THE COMMITTEE ON THE JUDICIARY UNITED STATES SENATE — APRIL 10, 2018
JOHN THUNE, South Dakota, Chairman; ROGER F. WICKER, Mississippi; BILL NELSON, Florida, Ranking...

Step 1: Approach — Semantic Partitioning & Map-Reduce

- **Context Limits & Partitioning:** Long transcripts exceed token limits and span diverse topics, requiring structure-aware chunking.
- **Speaker-Turn Parsing:** Rule-based parser separates 903 speaker turns without LLM calls, preserving dialogue context.
- **Hierarchical Map-Reduce:**
  - *Map:* 55 chunks $\rightarrow$ 55 segment summaries ($\approx$ 150 words each).
  - *Collapse:* 55 segment summaries $\rightarrow$ 3 condensed accounts ($\approx$ 200 words each).
  - *Reduce:* 3 condensed accounts $\rightarrow$ 1 hearing summary (264 words, 359 tokens).
- **Total Calls:** 59 model calls for 243k words (evaluated on Qwen 3.5-4B).

Architecture: Map-Reduce Summarization Pipeline



Qwen3.5-4B, temperature 0.1; every call logged (29,264 calls, prompt `hearing-mapreduce-1.0`).

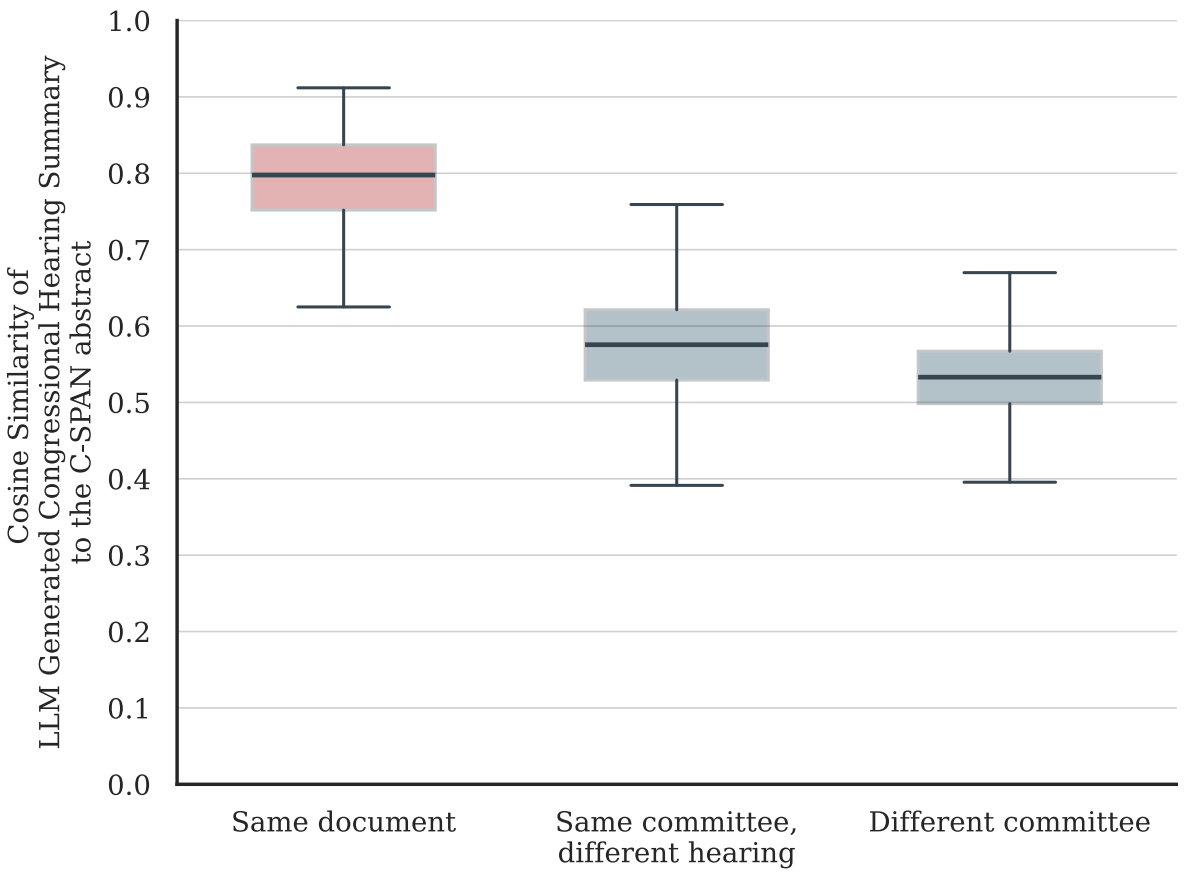
Step 1: Validation Against Independent Human Abstracts

- **Ground-Truth Benchmark:** Evaluated against 3,812 human-written C-SPAN editorial abstracts using *bge-large-en-v1.5* embeddings.
- **Discriminant Validity Test:** Cosine similarity compared across three mutually exclusive tiers:

Comparison	Assessment Objective	Pairs	Median Sim.
Same hearing	Correspondence to independent human account	351	**0.798**
Diff. hearing, shared comm.	Baseline for recurring committee subject matter	94,304	**0.576**
Hearings, no shared comm.	Broad cross-congressional baseline similarity	1,243,357	**0.533**

- **Top-1 Retrieval:** Correct human abstract ranked #1 among all 3,812 candidates for **61.3%** of hearings.

Step 1: Similarity Distribution



Note: each observation is one pair — the generated summary of a hearing against one C-SPAN abstract. 351 science-committee hearings, 1999–2025, matched to C-SPAN by date, chamber, title and witnesses, compared against 3,812 abstracts from 47 committees. LLM model used: Qwen/Qwen3.5-4B. Embeddings: BAAI/bge-large-en-v1.5 at 512 tokens, no truncation.

Step 1: Three-Tier Research Outputs

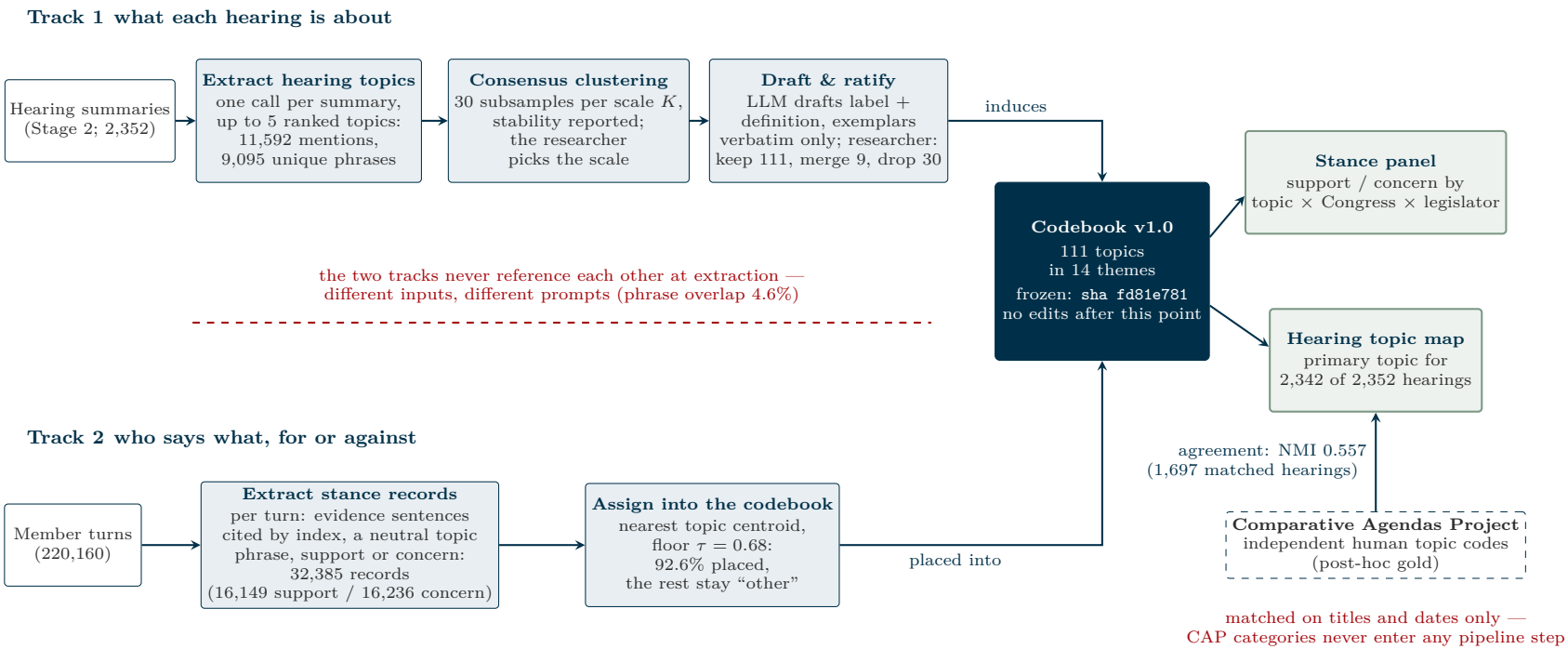
- **Traceable Asset Generation:** Releasing all intermediate stages for auditability:

Output Tier	Content	Research Utility
Parsed Records	Cleaned turns, speaker labels, and chunk bounds	Trace raw text; inspect turn boundaries
Segment Summaries	Intermediate outputs before final synthesis	Identify if details were lost in mapping
Hearing Summary Table	2,352 hearing rows with metadata and runtime	Extract topics; compare accounts at scale

Step 2: Two-Track Topic and Stance Extraction

- **Beyond Bag-of-Words:** LLMs identify semantic concepts directly from dialogue without LDA topic mixing or vocabulary thresholds.
- **Track 1 (Hearing Subjects):** Model extracts up to 5 neutral policy phrases per summary (*"Facebook data privacy"*, *"Honest Ads Act"*, *"GDPR-style controls"*).
- **Track 2 (Legislator Stances):** Cleans 507 legislator speaking turns (1,696 sentences); extracts neutral topic, stance (*support* vs. *concern*), and sentence citations.
- **Evidence Linking:** Links statements directly to numbered sentence citations for transparent human review.

Pipeline: Member Turn Topic & Stance Extraction

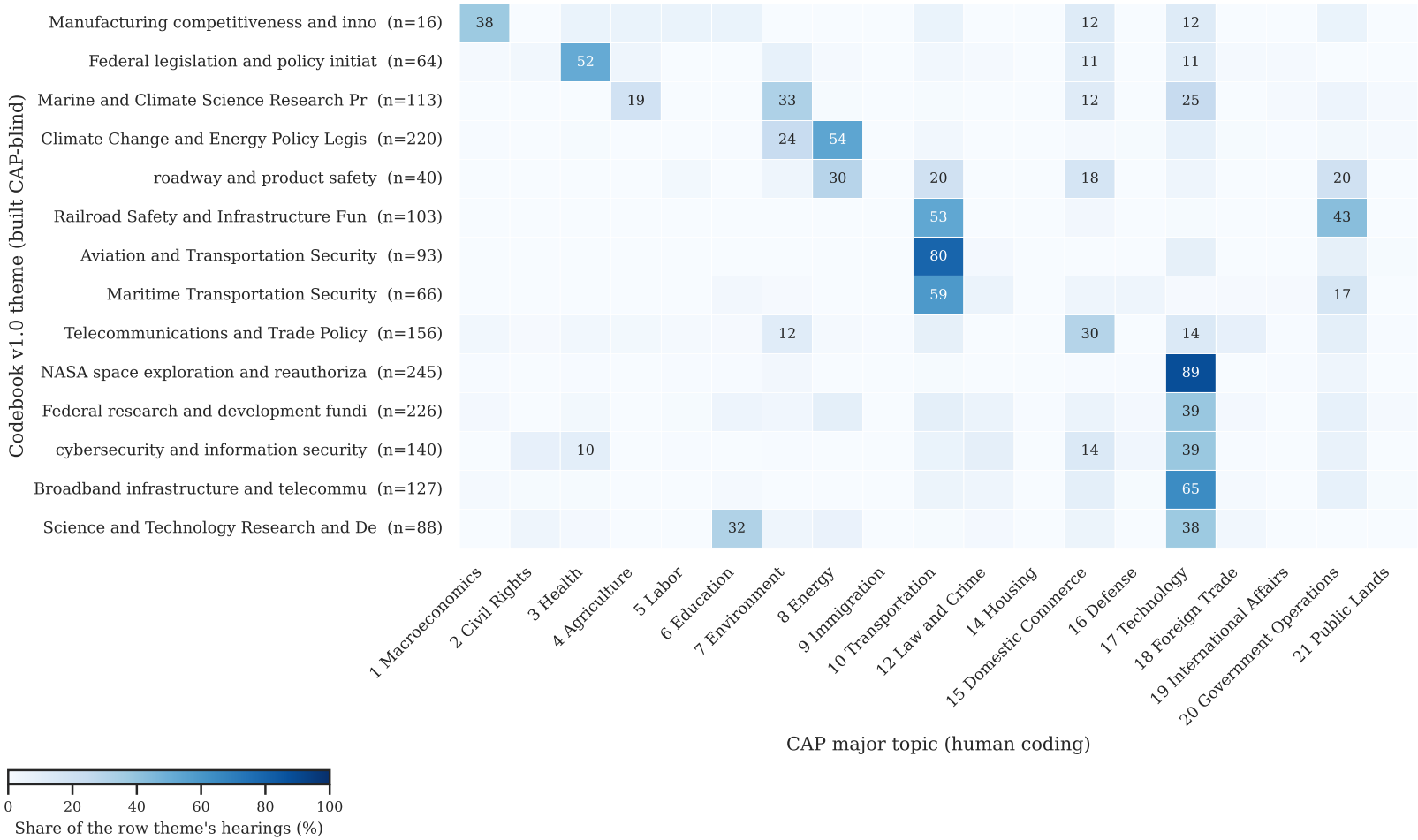


Qwen3.5-4B, temperature 0.1; prompts `summary-topics-0.2-dev` (Track 1) and `stance-extraction-0.3-dev` (Track 2); every call logged.

Step 2: Validation Against Comparative Agendas Project (CAP)

- **Benchmark Sample:** 1,697 matched hearings with official human-assigned CAP codes.
- **Phrase Correspondence:** Human-assigned CAP subtopic ranks #1 for **27.0%** of hearings.
- **Domain Alignment:** Cross-tabulation shows strong correspondence (e.g., 88.6% of NASA theme hearings align with CAP Technology domain).

Figure: CAP Category Crosswalk Heatmap



Note: 1,697 matched hearings, House Science and Senate Commerce committees, 1996–2021. Similarity is cosine similarity in the BAAI/bge-large-en-v1.5 embedding space between the hearing's extracted topic and each CAP category description; the taxonomy is the frozen Track-1 codebook v1.0 (111 topics, 14 themes). Cell numbers are row shares of 10% or more; rows ordered by modal CAP destination.

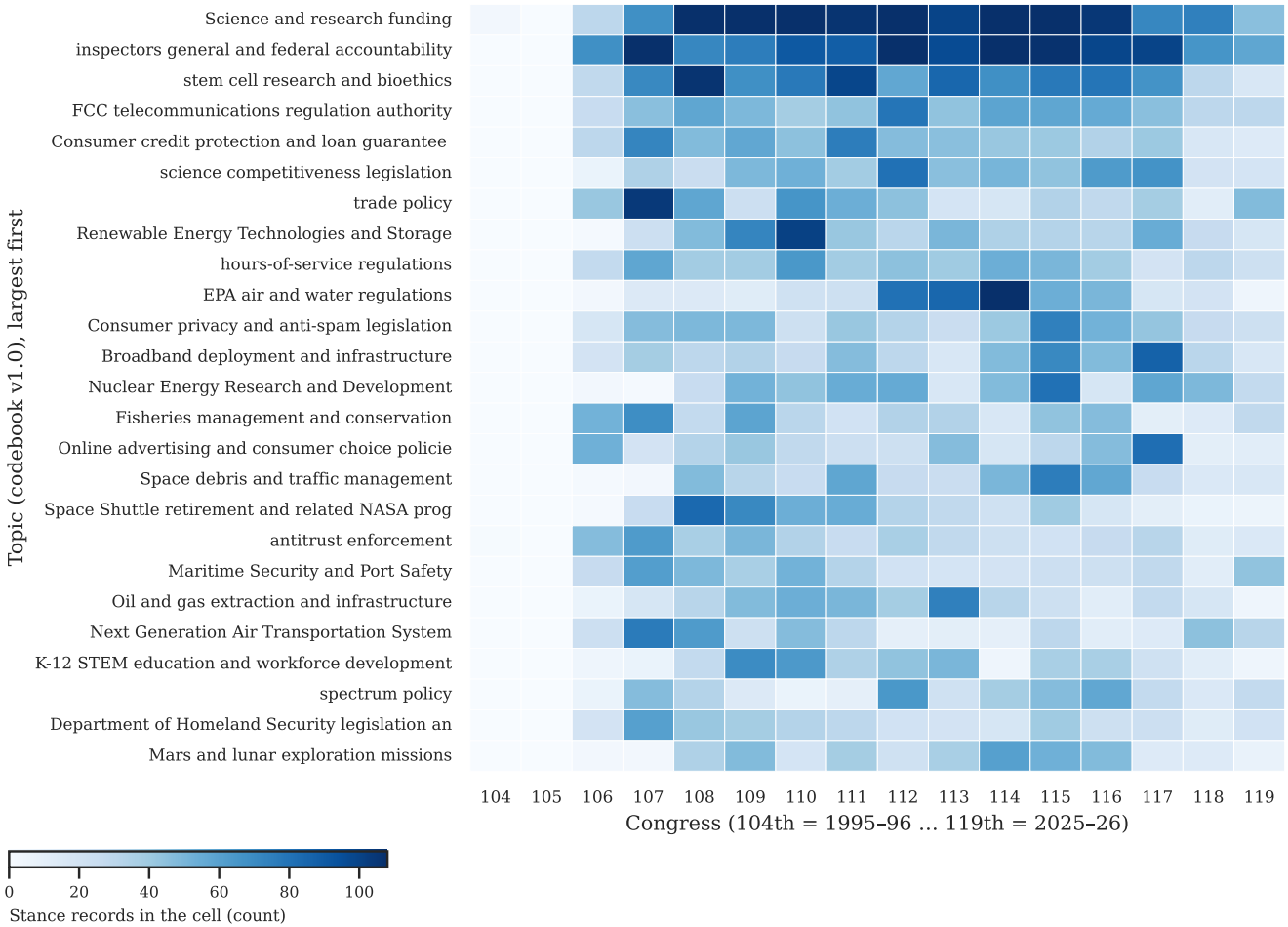
Step 2: Longitudinal Stance Panels & Polarization

- **Corpus Totals:** 32,385 stance records across 2,352 hearings.
- **Topic-by-Congress Aggregation:** Panel measuring volume and net stance:

$$\text{Net Stance} = \frac{\text{Support Records} - \text{Concern Records}}{\text{Support Records} + \text{Concern Records}} \in [-1, +1]$$

- **Example (Consumer Privacy in 115th Congress):** 17 hearings, 34 support, 21 concern records $\rightarrow$ Net Stance $= +0.236$.

Figure: Congressional Attention Heatmap



Note: 29,979 legislator stance records (prompt stance-extraction-0.3-dev) placed on the frozen Track-1 codebook v1.0; House Science and Senate Commerce hearings, 104th–119th Congress (1996–2025). A record is one member position: evidence sentences, a topic, and support or concern. Color depth is the number of stance records a topic drew in that Congress.

Figure: Net Stance Polarization Heatmap



Note: 29,979 legislator stance records (prompt stance-extraction-0.3-dev) placed on the frozen Track-1 codebook v1.0; House Science and Senate Commerce hearings, 104th–119th Congress (1996–2025). A record is one member position: evidence sentences, a topic, and support or concern. Red = concern-dominant, navy = support-dominant, gray = balanced; cells with fewer than 5 records are blank.

Step 3: Measuring Cross-Domain Topic Relevance

- **The Empirical Goal:** Connect Congressional policy topics to publicly traded firms via SEC 10-K Item 1 (Business Description) disclosures.
- **Corpus Scale:** 21,966 filings for 1,061 S&P 500 firms (6,306,793 retained sentences after boilerplate removal).
- **Two-Layer Architecture:**
  - *Layer 1 (Deterministic Embedding Rail):* Sentence-level cosine similarity to fixed topic centroids (top 5 sentences averaged).
  - *Layer 2 (LLM Sentence Tagging):* Qwen two-pass screening and confirmation of directional framing (*Opportunity*, *Threat*, *Neutral*).

Step 3: Similarity-Weighted Directional Scores

- **Mathematical Formulation:**
  - Let $w_{sk} = \max(0, \cos(\mathbf{e}_s, \mathbf{v}_k))$ be the topic similarity weight of sentence s.
  - Separate threat and opportunity indices (per 100 retained sentences):

$$\text{Threat}_{fkt} = \frac{100}{N_{ft}} \sum_{s \in S_{ft}} w_{sk} \, \mathbf{1}\{D_{sk} = T\}$$

$$\text{Opportunity}_{fkt} = \frac{100}{N_{ft}} \sum_{s \in S_{ft}} w_{sk} \, \mathbf{1}\{D_{sk} = O\}$$

- **Advantage:** Combines the volume of discussion with its semantic proximity to the policy topic without letting threats and opportunities cancel.

Step 3: Firm Evidence and Score Comparisons

- **Embedding Similarity (Snapshot preceding April 2018 hearing):**
  - *Facebook (78 sentences):* Raw similarity = 0.759, Standardized = 4.879 (Sentence 48: subject to privacy & data protection laws).
  - *ResMed (484 sentences):* Raw similarity = 0.770, Standardized = 2.443 (Sentence 459: laws governing privacy & medical data).
- **Unweighted vs. Similarity-Weighted Scores:**

Firm	Unw. Threat	Wt. Threat	Unw. Opp	Wt. Opp
Total System Services	24.14%	**16.03**	0.00%	**0.00**
Facebook	19.23%	**13.65**	0.00%	**0.00**
Paycom Software	1.67%	**1.25**	2.67%	**1.46**
Fidelity National Info (FIS)	9.27%	**6.16**	1.16%	**0.81**

Step 3: Sentence Evidence for Directional Labels

- **Sentence-Level Audit:** Retaining original disclosure sentences behind model classifications:

Firm	Sent.	Label	Original 10-K Disclosure Sentence
Total System Services	48	**Threat**	In many jurisdictions consumers must be notified in the event of a data breach, and such notification requirements continue to increase in scope and cost.
Facebook	55	**Threat**	The GDPR will include operational requirements... and will include significant penalties for non-compliance.
Paycom Software	263	**Opportunity**	We voluntarily obtain third-party security examinations relating to security and data privacy in accordance with SSAE No. 16...
FIS	217	**Opportunity**	The Company has adopted a comprehensive global privacy program to assess and manage these evolving risks...

Figure: Privacy Measure Agreement Across Quintiles

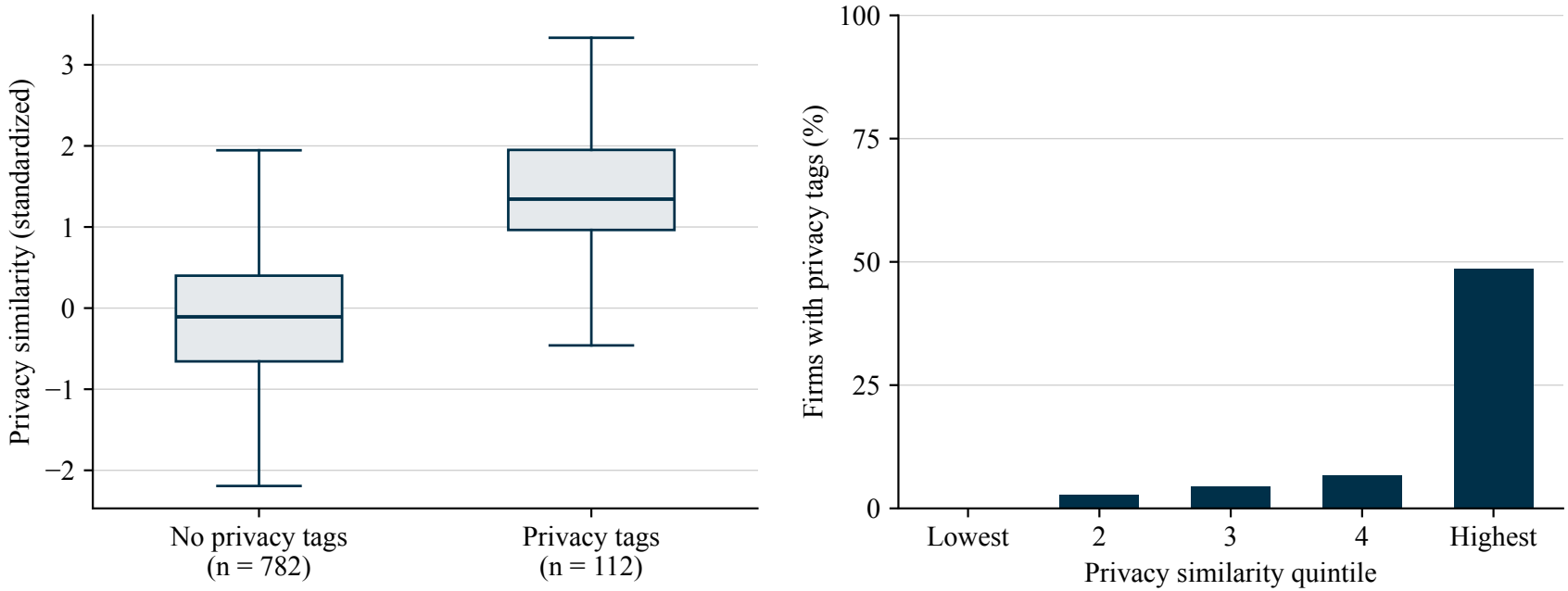
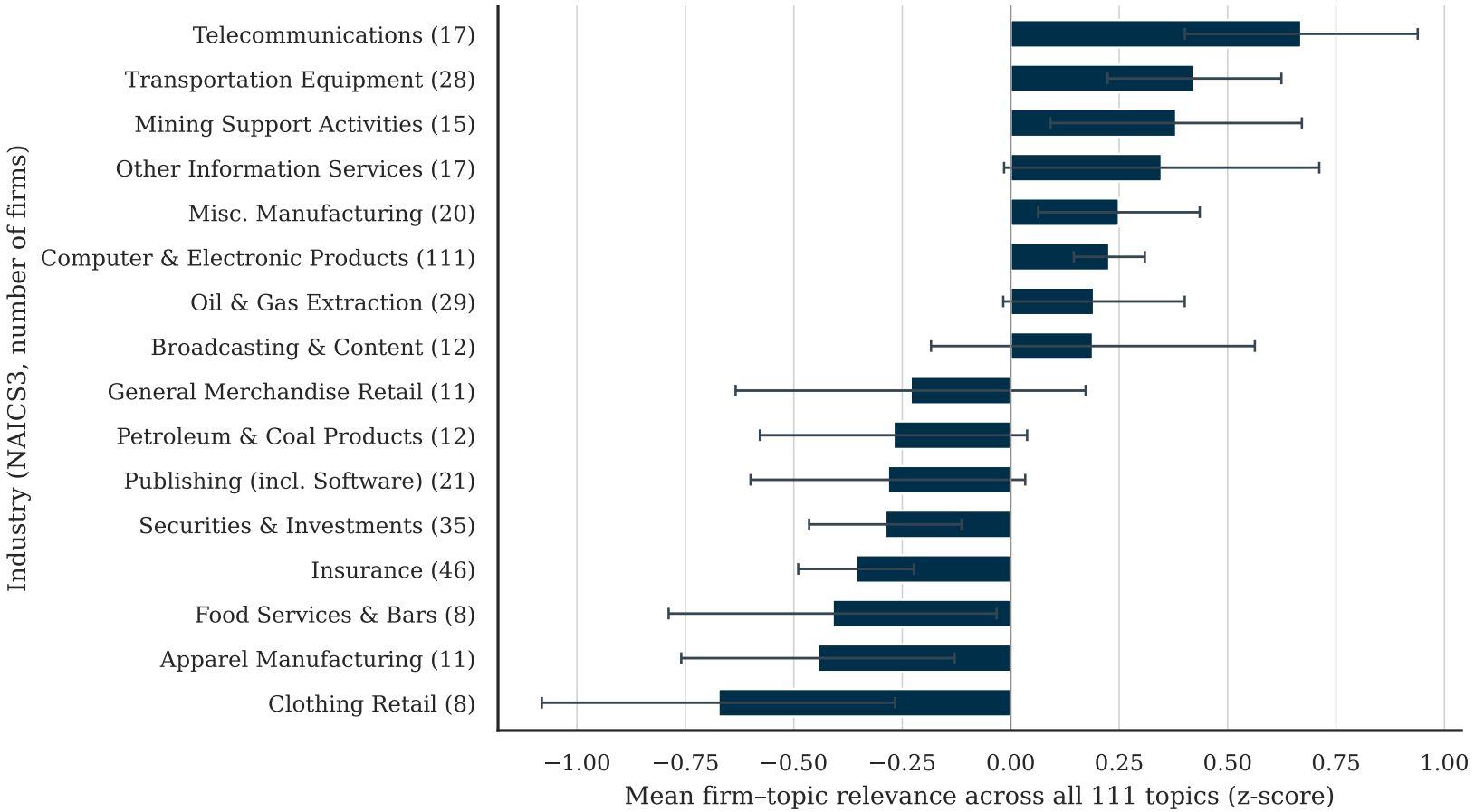
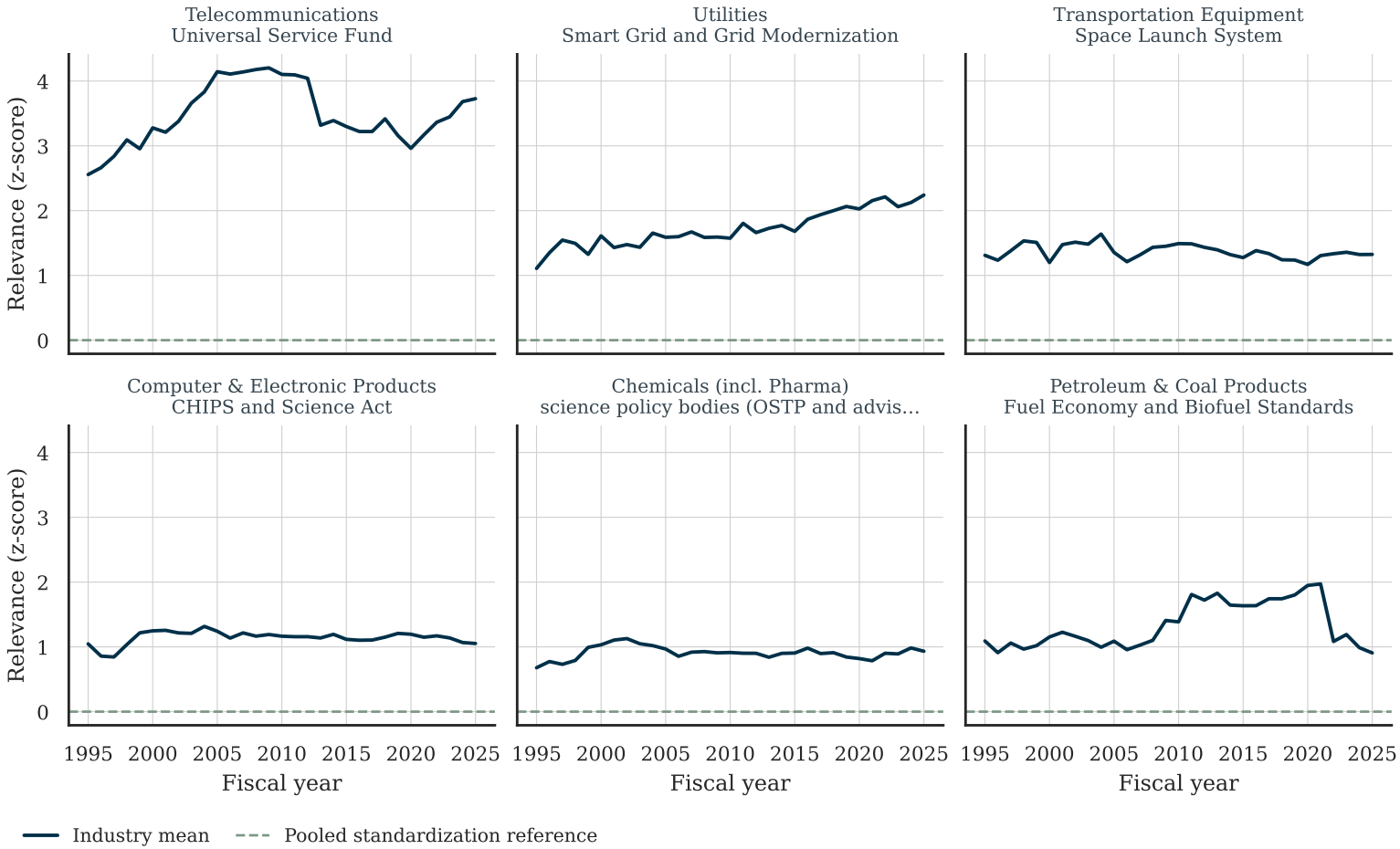


Figure: Industry-Wide Policy Exposure



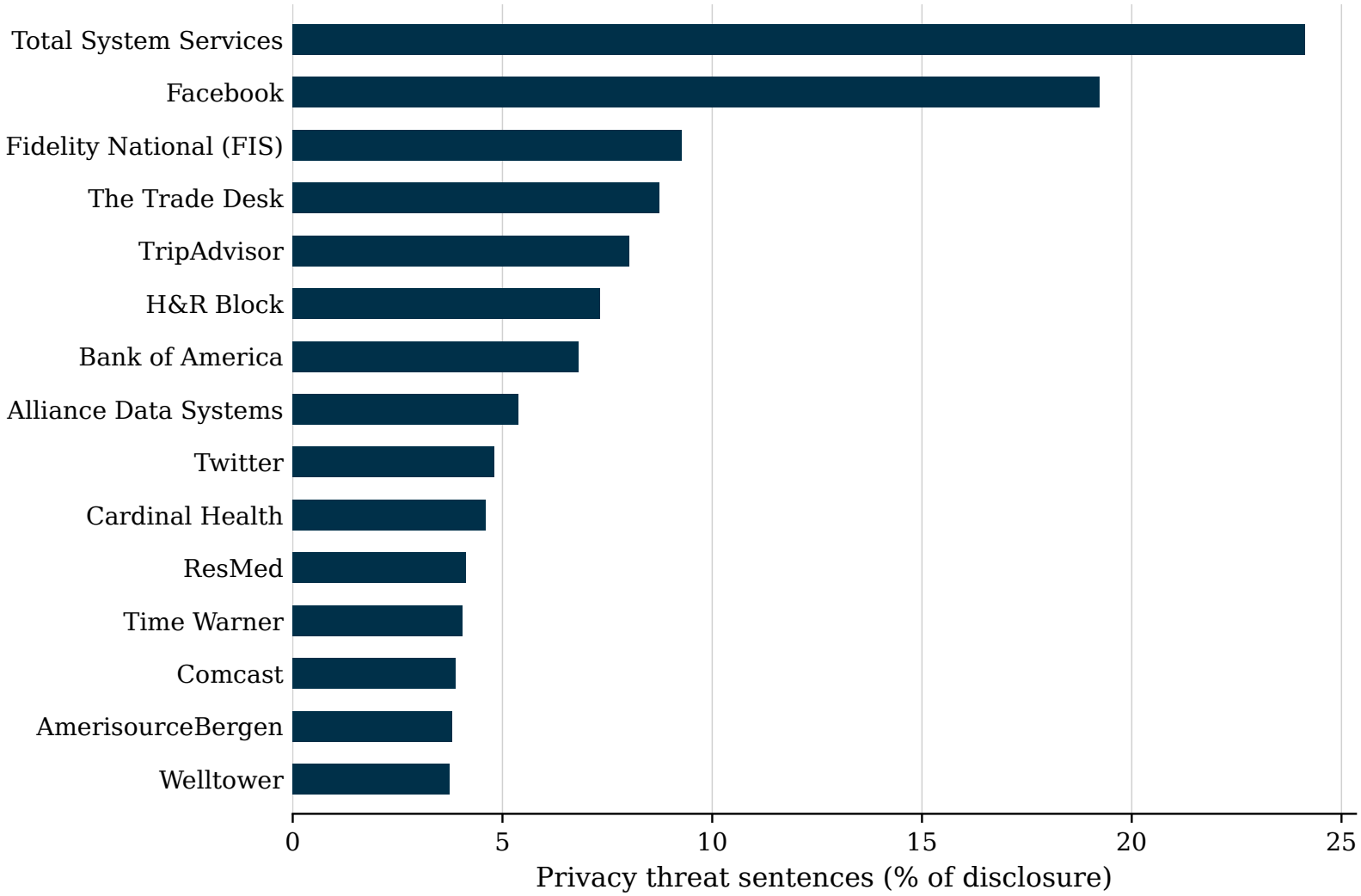
Note: The eight most and eight least agenda-exposed NAICS3 industries with at least eight S&P 500 firms, 1992–2026. Each firm contributes one observation: its mean standardized relevance (z within topic and filing-length decile) across all 111 frozen codebook topics and all of its 10-K filings. Bars are industry means; whiskers are 95% confidence intervals across firms. Positive values: the industry's Item 1 language sits closer to the House Science and Senate Commerce hearing agenda than the typical S&P 500 firm's.

Figure: Longitudinal Industry Disclosure Trends



Note: Yearly industry-mean standardized relevance (z within topic and filing-length decile) for six industry–topic pairs; each panel's topic is the one its industry scores highest on over the whole period. The dashed line at zero is the pooled topic-and-length standardization reference, not a yearly industry mean. S&P 500 10-K Item 1 filings, fiscal years 1995–2025.

Figure: Leading Privacy Threat and Opportunity Ranks



Step 3: Empirical Assets of Topic Relevance

- **Multi-Level Panel Assets:**

Output Tier	Content	Research Utility
Prepared Sentences	6.3M 10-K sentences with metadata	Retrieve raw text; test preprocessing
Topic Vectors	111 representations combining text + evidence	Fixed anchor representation across years
Corpus Similarity	2.4M filing–topic pairs with top-5 citations	Measure proximity across S&P 500 firms
Weighted Panel	Threat & Opportunity indices (894 clean filings)	Econometric panel on corporate policy exposure

Part B

LLM Coding Agents for Research & PhD Teaching

Evaluating Current Discourses on LLMs in Research

- **Can AI generate novel theory or formulate and test hypotheses?**
  - Currently, no. But this is not what we should want.
- **Can LLMs effectively synthesize literature and map a field?**
  - Valuable for orienting new entrants, but often superficial for established scholars.
- **Can AI assist in data cleaning, structuring, and synthesis?**
  - Yes, [as shown in Part A].
- **Can LLMs facilitate the execution of technical research tasks?**
  - Yes, particularly in writing and optimizing code.
  - The critical challenge shifts from syntax generation to formulating research designs and processes.

Evolving the Data Pipeline: An Agent-Centric Approach

- **Transitioning to "Agent-First" Workflows:** Moving beyond simple chat interfaces to deploying autonomous agents to do the analytical tasks.
- **Understanding of the context:** While AI generates code, the researcher must possess a deep understanding of the codebase architecture and the overall data stack.
- **Navigating the Computational Environment:** Need to understand the packages, functionality and what the technology is doing.
- **Shifting Cognitive Load:** The responsibility of the researcher evolves from knowing syntax to architecting the analytical structure.

Implications for Doctoral Education

- **Enhanced Reproducibility:** LLMs enable the generation of cleaner, more robust code, which the LLM can audit.
- **Rethinking Doctoral Training:**
  - Historically, methodological training emphasized the mechanics of coding.
  - Shifting focus to teaching students how to develop analytical workflows.
- **The Limitations of Traditional Replication:** While replicating prior studies remains valuable, existing approaches focus on reproducibility.
  - LLMs allow for better documentation, in-process validation, human-readable documents, and clarification of the logic.

The Future of Research Outputs

Current fear of LLMs being able to "do research" better than people → It's already there.

- **Elevating Research Questions:** The premium shifts away from "shock arbitrage" toward important research questions.
  - Because LLMs can evaluate alternatives, the value is not the empirical contingencies. The value is the question and the data.
- **Transforming the Manuscript Format:** If empirical execution becomes easy, why retain dense, convoluted methods sections?
  - We may transition toward methods sections functioning as high-level executive summaries, with the granular methodology dynamically documented and accessible in public repositories.

Concluding Thoughts on Research Pipelines

- **LLMs as Implementation Engines:** They are exceptionally powerful tools for executing complex research designs.
- **The Era of Agents:** The debate has moved past simple chatbots; autonomous agents are already transforming commercial software development and will similarly impact academic research.
- **Iterative Refinement:** The value lies in using LLMs to continuously check, refine, and adapt code across diverse empirical scenarios.

Concluding Thoughts on Institutional Shifts

The Real Existential Question:

While many worry that AI will devalue traditional research skills, the more profound disruption is institutional:

- These technologies challenge the primacy of the static, "PDF-centric" publication model.
- The critical question is determining how to attribute intellectual credit, manage the peer review process, and document scientific knowledge in a dynamic landscape.

Part C

LLM Coding Agents for Teaching

LLM Coding Agents and the Future of Teaching

- **Productivity & Adoption:** This is an undeniably positive shift that increases productivity; everyone should learn and adopt these tools.
- **The Wrong Focus:** The discourse is often fixated on the wrong things (theorizing, long-form writing, etc. were just testing proxies).
- **Two Core University Functions:** Universities perform two intertwined roles: **Teaching/Knowledge** and **Evaluation**.
  - We were never exceptionally good at evaluation, but knowledge and evaluation were historically highly correlated.
  - Difficult exams with trick questions or elaborate essays meant students had to master knowledge to succeed.
- **Diminishing Correlation:** LLMs make learning easier, but make evaluating learning with traditional methods far harder.
- **The Real Disruption:** Don't worry about direct teaching decline undermining universities; worry about employers and companies performing evaluation directly.

Potential Solutions: Real Problem-Solving

- **Evaluation Through Building:** Test by building and solving real problems rather than evaluation through repetition or trick questions. Building with LLMs enables students to accomplish far more than ever before.
- **Current Misalignment:** Those who could benefit most from AI often avoid it, while others overuse tokens prognosticating hype.
- **A Practical Experiment:** Rather than managing central token licenses, give staff and students $20/month for their own agent models to use across personal and professional tasks to gain genuine proficiency.
- **Focus on Immediate Problems:** Those who solve immediate, concrete problems with these tools consistently outperform those chasing grandiose theoretical claims.

Thank You!

Questions & Discussion